ASSIGNMENT-04 (LINUX)

1. List all meta characters used in the Linux shell and explain their purpose with examples.

* : Matches or Wildcard for multiple characters. Example: ls *.txt

? : Matches or Wildcard for a single character. Example: ls file?.txt

| : Pipe operator used to pass output of one command to another. Example: ls | grep txt

> : Redirects output to a file. Example: ls > files.txt

>> : Appends output to a file instead of overwriting. Example: echo "Hello" >> file.txt

$: Used for variable expansion. Example: echo $HOME

2. Write a command that uses the wildcard `*` to list all `.txt` files in a directory.

naman@naman-virtualbox:~/shell_scripts$ ls *.txt

3. Write a command that uses the wildcard `?` to list files that have exactly five characters in their name.

naman@naman-virtualbox:~/shell_scripts$ ls ?????

4. Demonstrate output redirection by sending the output of `ls` into a file named `filelist.txt`.

naman@naman-virtualbox:~/shell_scripts$ ls > file.txt

5. Write a command that appends output of the `date` command to a file called `log.txt`.

naman@naman-virtualbox:~/shell_scripts$ date >> log.txt

6. Show how to pipe (`|`) the output of `ls` into the `grep` command to find files containing the word `test`.

naman@naman-virtualbox:~/shell_scripts$ ls | grep 'test'

7. Write a command that uses command substitution `$()` to display the current date inside an echo statement.

naman@naman-virtualbox:~/shell_scripts$ echo “$(date)”

8. Demonstrate how to use logical operator `&&` to run two commands only if the first command succeeds.

naman@naman-virtualbox:~/shell_scripts$ echo “Hello Ajit” > ajit.txt && cat ajit.txt

9. Write a command that runs `sleep 60` in the background.

naman@naman-virtualbox:~/shell_scripts$ sleep 60 &

10. Show the command used to list all background jobs currently running in the shell.

naman@naman-virtualbox:~/shell_scripts$ jobs

11. Write a command to bring a background job to the foreground.

naman@naman-virtualbox:~/shell_scripts$ fg 7078

12. Display the last 10 commands executed in the shell using the history command.

naman@naman-virtualbox:~/shell_scripts$ history | tail –n 10

13. Write a command to execute the last command again using shell history.

naman@naman-virtualbox:~/shell_scripts$ history | tail -n 1

OR

naman@naman-virtualbox:~/shell_scripts$!!

14. Execute the 5th command from command history using the appropriate syntax.

naman@naman-virtualbox:~/shell_scripts$ History | head -n 5 > cmd.txt | tail -n 1 cmd.txt | ./cmd.txt

OR

naman@naman-virtualbox:~/shell_scripts$!5

15. Create a file named example.txt and change its ownership to user `user1`.

```
naman@naman-virtualbox:~/shell_scripts$ touch exp.txt
naman@naman-virtualbox:~/shell_scripts$ sudo chown user1 example.txt
```

16. Write a command to change the owner and group of `example.txt` to `user1` and `group1`.

```
naman@naman-virtualbox:~/shell_scripts$  chown user1:group1 example.txt
```

17. Create a file named cdac.txt and assign permissions such that the owner has rwx, group has rx, and others have no permissions.

```
naman@naman-virtualbox:~/shell_scripts$ touch cdac.txt
naman@naman-virtualbox:~/shell_scripts$ chmod u+x cdac.txt
naman@naman-virtualbox:~/shell_scripts$ chmod g-w cdac.txt
naman@naman-virtualbox:~/shell_scripts$ chmod g+x cdac.txt
naman@naman-virtualbox:~/shell_scripts$ chmod o-r cdac.txt
```

18. Write a command to view the Access Control List (ACL) of a file named `file1`.

```
naman@naman-virtualbox:~/shell_scripts$ getfacl file1
```

19. Write a command to give user `user2` read and write permissions on `file1` using ACL.

```
naman@naman-virtualbox:~/shell_scripts$ setfacl -m u:user2:rw file1
```

20. Write a command to remove ACL permissions for user `user2` from `file1`.

```
naman@naman-virtualbox:~/shell_scripts$ setfacl -x u:user2 file1
```

21. Write a command to mount a filesystem from `/dev/sdb1` to `/mnt/external`.

```
naman@naman-virtualbox:~/shell_scripts$ mount /dev/sdb1 /mnt/external
```

22. Write a command to unmount a mounted filesystem from `/mnt/external`.

```
naman@naman-virtualbox:~/shell_scripts$ umount /mnt/external
```

23. Use `grep` to search for the word "linux" in file.txt ignoring case sensitivity.

```
naman@naman-virtualbox:~/shell_scripts$ grep -i "linux" file.txt
```

24. Write a `grep` command to count how many times the word "error" appears in a file.

naman@naman-virtualbox:~/shell_scripts$ grep -c "error" file.txt

25. Write a `find` command to locate all `.txt` files in the current directory and its subdirectories.

naman@naman-virtualbox:~/shell_scripts$ find . -name "*.txt"